MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

STUDY GUIDE UNFCCC-COP

Agenda 1 : Assessing the Impact of Climate Change on Small Island States

Agenda 2 : Advancing Climate Action in Global Food Security.

Letter from R&D

Dear Delegates,

The Department of Research & Development of MUNSociety MPSTME welcomes you to the 11th edition of Mumbai MUN! Our department upholds a proud tradition of facilitating meaningful discourse on national and international matters, both past and present.

We are primarily tasked with formulating agendas and preparing comprehensive study guides. This year, we have crafted comprehensive and engaging guides designed to be valuable resources for all delegates.

This year's theme, "Redefine the Norm," challenges us to think beyond conventional solutions and embrace innovative approaches to global challenges. In a rapidly evolving world, traditional methods often fall short of addressing contemporary issues. We believe that transformative change requires the courage to question established practices and the creativity to envision alternative pathways.

With this theme in mind, we have thoroughly picked agendas and committees that demand fresh perspectives, challenge conventional thought, and encourage innovative approaches to problem-solving. In this study guide, you will find well-researched background information to help you prepare for your committee sessions.

We, as facilitators, aim to provide you with the intellectual arsenal to combat any challenges that you might face and make the most of your Mumbai MUN experience.

We look forward to your active participation and contributions during Mumbai MUN and are confident that, together, we can make this conference a memorable and enriching experience for all.

Best wishes,
Department of Research & Development,
MUNSociety MPSTME

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Note: The QARMAs will be released pre-conference, post consultation with the Executive Board

COMMITTEE OVERVIEW

The United Nations Framework Convention on Climate Change (UNFCCC) is the United Nations Entity established in the year 1992 which is responsible for supporting the global response to the threat of climate change. It has 198 parties and is the parent treaty of the 2015 Paris Agreement and 1997 Kyoto Protocol.

The main work of the UNFCCC is to stabilise greenhouse gas concentrations in the atmosphere to levels that won't affect the climate system dangerously as well as promoting sustainable development and the natural growth of ecosystems. The main aim of the Paris Agreement is to keep the global average temperature rise this century as close as possible to 1.5 degrees Celsius above pre-industrial levels. The Kyoto Protocol similarly has the goal of ensuring that the industrialised countries and economies in transition limit and reduce the emission of greenhouse gases by their agreed targets.[1]

The Conference of Parties (COP) is the supreme decision-making body of the UNFCCC that comes together to review implementations of the Convention and any other legal instrument that the COP adopts and implements solution-related ideologies that could help a smooth and effective implementation of the Convention and better the situation worldwide. The COP meets every year to assess the effects taken by the Parties regarding the situation and check on the progress embarked on by them in achieving the desired goals.

MANDATE

Within the UNFCCC-COP mandate, the Committee is responsible for having an exchange of information–without any prejudice–regarding actions relating to mitigation and adaptation to assist Parties in continuing to develop effective and appropriate responses to climate change. Along with this, there is a clear discussion over the measures and policies that have been implemented by each of their respective governments, confirming their stand and determination towards the goal. The conferences are held to ensure that each party is represented and extend a helping hand in financing the presence of the developing country's governmental experts. This leads to a better understanding of the situation globally and out of the discussions held, emerges viable solutions that can be adapted to make changes towards climate change.

[1] un. (n.d.). *UNFCCC*. UNFCCC.int. https://unfccc.int/

Agenda 1: Assessing the Impact of Climate Change on Small Island States

AGENDA OVERVIEW

This agenda aims to understand the root issues faced by the inhabitants of the Small Island Developing States as a result of climate change and the barriers restricting them from achieving sustainability and resilience. Climate change has innumerable impacts such as natural calamities, ecosystem and biodiversity endangerment, and infrastructural destruction, which leads to the deterioration of the environment. With several nations being deeply affected by this cause–in terms of health, safety and economy–effective measures must be taken into action and a properly established framework should be designed with the convention of reducing the severe impacts of climate change.

INTRODUCTION TO SMALL ISLAND DEVELOPING STATES

Small Island Developing States (SIDS) is the terminology designated to a group of 39 distinct states and 18 Associate Members of United Nations regional commissions that are economically, environmentally and socially vulnerable. They were recognised by the United Nations at the 1992 Conference on Environment and Development conducted in Rio De Janeiro, Brazil.[2] From the environmental perspective, these developing regions face the most catastrophic natural calamities and extreme weather events due to their geography. In recent times, Cyclone Winston and Cycle Harold hit Fiji and Vanuatu respectively, causing infrastructure damage to a great extent. The ocean-to-land mass ratio is 28:1, making the SIDS heavily dependent on the ocean for their livelihood. They contribute almost less than 1 percent of global greenhouse gas emissions yet are the most impacted by climate change. The impacts of climate change can be seen through Infrastructure damage, deteriorating human health, environmental degradation, occasional coastal floodings etc. This situation persists due to restrictions such as insufficient monetary aid, lack of public comprehension, outlying locations, and inefficient government support. The UN has instigated several Programmes of Action in favour of SIDS such as the Barbados Programme of Action, and SAMOA Pathway.

Adverse Effect of Climate Change on SIDS
Source: OPEC Fund

CAUSES

Meteorological Factors

Global Warming is a threat to humanity and prevails to prove its fatality by wrecking the biosphere. Although not a heavy contributor, SIDS face the wrath of global warming through an immense rise in sea levels, ocean warming, and catastrophic temperature surges. Ocean warming jeopardizes the aquatic ecosystem, leading to coral bleaching and negatively affecting the health of marine species. Ocean acidification poses a hazard to the tourism and fisheries industry and reduces the protection imparted by the coral reefs during storm surges. According to reports, in the last few months of

[2] United Nations. (n.d.). Conferences | Small Island Developing States. United Nations. https://www.un.org/en/conferences/small-islands#:~:text=Small%20Island%20Developing%20States%20(SIDS,in%20Rio%20de%20Janeiro%2C%20Brazil.

2023, it was noted that the average surface temperature escalated by 1.7°C in the SIDS than when perceived in the 1951-1980 reference period, exceeding the desired limit of 1.5°C set by the Paris Agreement This increase corresponds to higher drought risks and hinders the availability of freshwater, compromising local flora and fauna. Without efforts to limit the temperature increase, the prevalence of disasters will continue to grow, and survival will become increasingly challenging.

Carbon Emissions

One of the major catalysts to climate change is the emission of greenhouse gases, particularly carbon dioxide. Although SIDS contribute less than 1 percent of global carbon emissions, they bear the brunt of the consequences.[3] Carbon emissions severely affect the ecosystem as it hampers the rainfall patterns leading to large-scale flooding, and calamities such as droughts and rising sea levels. It also alters the acidic levels present in water harming marine species and hampering the ability of waterbodies to absorb carbon emissions from the atmosphere.

Aquatic Influences

The SIDS are characterised by their low-lying terrain, which makes them highly susceptible to flooding. Nations in the SIDS, with land less than five metres above sea level, face even more deplorable conditions, as they are at risk of being wiped out entirely. For instance, the Maldives, where over 80 percent of the land is less than a metre above sea level, faces the threat of extinction due to rising sea levels. As sea levels rise, the frequency and intensity of storms and cyclones increase, further endangering the inhabitants and disrupting their livelihoods.

External Dependency

The economies of SIDS are heavily reliant on the ocean and its resources. As reported by the UNDP in 2017, SIDS consist of 96.5 percent ocean and only 3.5 percent land, highlighting their dependence on marine resources for their livelihood. The tourism and fishing industries account for up to half of the GDP in countries like Antigua and the Maldives. However, the overexploitation of these resources is destabilising ecosystems, making them more vulnerable to climate change. With limited alternative sources of livelihood or economic resilience, SIDS are particularly exposed to climate-related disasters. Additionally, SIDS face some of the highest electricity costs in the world due to their reliance on imported fossil fuels. This further drains financial resources, preventing investment in sustainable alternatives that could mitigate the impact of climate change.

IMPACTS

Lack of Freshwater Resources

Water is the most vital element for any species, yet climate change poses a severe threat to freshwater resources in SIDS. UNESCO reports that 71 percent of the SIDS face the danger of water shortage which even goes up to the bar of 91 percent in regions with the lowest altitude. The water so far available is also contaminated by the result of climate change in the form of seawater, mud and sewage–usually as a consequence of any natural disaster like storms, floods, or tsunamis–which flows into the freshwater bodies. The groundwater is also not safe from salinity as saltwater intrudes into it and because of that, there is a risk of 73 per cent of groundwater pollution. Lack of groundwater can also be tagged as a drought condition as this will

[3] Small island developing states. UNDP Climate Promise. (2019, July 31). https://climatepromise.undp.org/research-and-reports/snapshot-small-island-developing-states

not only affect humankind but their only source of sustenance as well. A proper interdisciplinary approach is required to mitigate this gruesome situation.

Biodiversity Endangerment

Coral Bleaching

Coral reefs play a vital role in marine ecosystems by providing habitat for numerous species and protecting coastlines from erosion. Coral Bleaching can be noticed when the corals turn white or lose their vibrant colour tone as a consequence of the stress of climate change. When the corals are discoloured or disfigured, this, in turn, affects the ecosystem surrounding it. The decline in coral health may also reduce the appeal of scuba diving tourism if the aesthetic and ecological integrity of the reefs is not adequately restored. Furthermore, the weakening of coral reefs poses a substantial ecological risk, including a potential decline in oxygen production, which could have serious consequences for marine life.

Ocean Acidification

The CO_2 present in the atmosphere when absorbed by the ocean alters the pH baseline making it more acidic. This is characterised by ocean warming, affecting coral reef growth and hindering the ability of marine species to make and maintain their shells–like shellfish, and sea urchins. Many of the SIDS are built from carbonate skeletal material which was accumulated from the dead marine calcifiers such as corals and calcifying algae. The increase in acidity of the water restricts the endurance of these structural reefs as calcium carbonate gets dissolved, endangering the regions.

Heat Waves

SIDS experience both terrestrial and marine heat waves. The higher the temperature, the more it becomes a threat to The atmosphere. Heatwaves give rise to hurricanes and destroy corals and mangroves which are essential in an ecosystem. Marine heatwaves primarily lead to ocean warming, which in turn drives the destruction of marine life. Similarly, the human body can only tolerate certain levels of heat, and with rising temperatures, it is expected that heat-related deaths will increase.

Health Disorders

Ciguatera Poisoning

Warm sea temperatures and the high frequency of cyclones damage coral reefs, which ultimately results in the growth of microalgae containing ciguatoxin. When fish ingest this microalgae, the toxins accumulate in their tissues. As SIDS rely on the ocean for their primary diet, consumption of these contaminated fish leads to ciguatera poisoning. There are no indications of the toxin's presence in fish without laboratory testing, making consumers susceptible to the disease. Over the past 20 years, the incidence has increased by up to 60 percent, with Pacific SIDS reporting approximately 500,000 cases annually. Kiribati is the most affected, with an incidence rate of 312 per 100,000 population per year. If not addressed properly, this poses a significant threat to the fishing industry, food security, and the economy.

Non-Communicable Diseases

Climate change has the potential to increase the incidence of infectious diseases, including malaria, dengue fever, and diarrheal illnesses. Furthermore, recent data reveal a significant rise in mortality rates associated with noncommunicable diseases.[4] There has also been strong evidence relating this situation to extreme weather conditions like heat waves, water insecurity as well unhealthy diets followed by the citizens. In Over 80 per cent of SIDS, more than one in six adults die prematurely from Non-Communicable Diseases. SIDS have one of the highest rates of risk of dying prematurely from any major NCDs, which are cancer, chronic lung illnesses, diabetes, and cardiovascular diseases. Beyond the impact on physical health, the stress caused by environmental destruction significantly undermines mental health, adversely affecting the well-being of the population.

Industrial loss

With their geographical locations surrounded by water and isolated from the congested world, Small Island Developing States (SIDS) attract many tourists. These regions rely heavily on tourism for income and livelihoods, making visitor engagement crucial to their economies. However, climate change and the consequent rise in natural disasters have led to significant losses for tourism-dependent nations. In The Bahamas, where 52.2 percent of jobs are supported by the tourism, the impact of five major hurricanes in recent years has altered visitors' perceptions, resulting in a decline in tourism-generated revenue.

Civic Displacement

Due to unpredictable meteorological conditions and crippling infrastructures, human existence in SIDS is increasingly threatened. Maldives–deeply dependent on tourism–had a survey in which over 90 per cent of their resorts reported beach erosion and 60 percent of them have undergone infrastructure damage. Moreover, sea level rise and coastal flooding cause significant damage to infrastructure, housing, and crops. In 2022, nearly 32 million people were displaced because of weather hazards which is 41 percent more than in 2008. As of 2024, about 300 Indigenous Guna families are being relocated from the Gardi Sugdub island to Panama as they fear the sea rise flooding. Many other members of SIDS are actively seeking refuge in other willing nations. Beyond the damage caused, the cost of restoring the environment to its original state ranges from 1 percent to 8 percent of a country's total GDP. In the context of already weak economic conditions, this financial burden can lead to long-term indebtedness, hindering the development process.

Damage brought about by Hurricanes
Source: The Conversation

BARRIERS

Financial Aid Scarcity

this globalised era, capital is essential for a nation's safety. This poses a significant challenge for SIDS. Although many of these nations fall under the middle-income status, they do not benefit from concessional funds, which are primarily allocated to low-income countries. Without these funds, SIDS struggle to recover effectively from climate change-related issues. Nations like Haiti and Comoros, classified as Least Developed Countries, heavily depend on

[4] World Health Organization. (n.d.-b). Climate Change and Noncommunicable Diseases in Small Island Developing States. World Health Organization. https://www.who.int/publications/m/item/climate-change-and-noncommunicable-diseases-in-small-island-developing-states

aid to sustain their economies and maintain well-being. According to UN reports, SIDS receive a minimal share of Official Development Assistance (ODA), hindering their growth and development. Additionally, the lack of technical aid is a major barrier to combating climate change. [5]

Along with a lack of aid, 40 per cent of the SIDS are suffering from unsustainable levels of debt and 70 per cent of the nations are not able to recover their economic losses. This again holds them back from investing in methods that could help them diminish the effects of climate change.

Insufficient Public Awareness

The challenges faced by Small Island Developing States remain largely unfamiliar to the broader public, despite public support being essential for overcoming these difficulties. SIDS have consistently voiced their need for assistance in addressing the adverse effects of climate change and in advocating for the importance of limiting global warming. Additionally, raising awareness among the populations of SIDS is crucial, which can be achieved through education, technology, and the dissemination of accurate information.

Resource Deficiencies

Apart from monetary and technical aid, there is a major requirement for resources that can fulfil sustainability goals. However, SIDS are heavily reliant on the ocean and have limited options for economic sustainability beyond imports. This dependency presents a significant barrier to economic growth and often drives these nations into debt. Overexploitation of the ocean resources can lead to scarcity in the future. The inability of to invest in renewable resources, due to constraints such as limited land availability, restricts their capacity for resource production and creates a substantial economic burden. Even the most basic groundwater resources are much lower than the required numbers which results in complicated situations during disasters. All these lead to a tougher post-recovery as the nation does not have adequate resources to counter the challenges.

Government Apathy

A significant concern with smaller governments is the increased risk of corruption, as there is less oversight from a larger population. Instances have been reported where financial aid designated for climate change resilience has been misappropriated for personal use by officials. Ensuring the fair and transparent allocation of public funds is essential, as any mismanagement could lead the nation into substantial debt and potentially trigger an economic collapse. Moreover, the lack of comprehensive policies addressing climate change poses a serious challenge, as governmental bodies should take the lead in formulating and implementing effective measures to mitigate the issue.[6]

Remoteness

While a remote location may offer a sense of safety and calm, the reality is often far from it. Greater distance results in less frequent transportation and connectivity, exacerbating

[5] United Nations. (n.d.-e). Finance for development for Small Island Developing States report - advance unedited | office of the high representative for the least developed countries, landlocked developing countries and Small Island Developing States. United Nations. https://www.un.org/ohrlls/news/finance-development-small-island-developing-states-report-advance-unedited

[6] Procurement and corruption in small island developing states. (n.d.-f). https://www.unodc.org/documents/corruption/Publications/2016/V1608451.pdf

the challenges faced by SIDS. Unpredictable weather conditions cause constant distress among the population. Remoteness also limits access to diverse resources, confining livelihoods largely to ocean-dependent activities.

This isolation makes disaster recovery more prolonged and costly, as immediate help from neighbouring areas is less accessible. Funds that could have been used for preventive measures are instead consumed by recovery efforts, further hindering the implementation of effective disaster preparedness strategies.

CASE STUDIES

Kiribati

Overview

The Republic of Kiribati is a collection of 32 coral atolls situated in the Pacific Ocean. It has a tropical climate with hot and humid temperatures throughout the year. The seasonal climate patterns have many variations generally due to the South Pacific and Intertropical Convergence Zones. It faced severe droughts in 1988-1989 and 2007-2009 and since then droughts and storms have been a particular cause of concern in the country.

Impacts

Kiribati, even while contributing least to the climate crisis, has experienced its worst effects. Kiribati's biodiversity has proven to be highly sensitive to heavy rainfall events drought and saline intrusion. The vulnerability and sustainability of groundwater resources under climate change have raised major concerns. These extremities in Kiribati are further enhanced by issues with governmental law enforcement forces. Reduced productivity of livestock, dwindling water resources and hence reduced agricultural productivity and a decline in water quality are the key impacts recognized so far. In 2010, the Government of Kiribati estimated that around 66 per cent of the total population is living under the poverty line or is at risk of falling below it. Combined with the nation's low food and water security, it has amplified its vulnerability to natural hazards driven by unpredictable climate phenomena.

Future Predictions

Many of the low-lying islands of Kiribati are predicted to be submerged as almost all islands in the country have a low-lying coastal zone. This is because the melting ice caps will lead to higher average sea levels over the coming decades. The frequency of coastal flooding will increase, leading not only to direct infrastructure damage but also to the decline and loss of cultivable lands. This will cause permanent destruction of ecosystems as wildlife will be unable to survive in such regions. Ocean acidification will disrupt coral reef replenishment pushing many marine species towards extinction which will affect the commercial fishing sector adversely. This pushes disparities to elevate beyond reason including poverty, inequality, gender disparity, social peace, and health. As the frequency of these incidents increases, human health will take a high toll contributing to malnutrition, lack of protection against diseases and overall well-being.[7]

Mitigation

Given Kiribati's vulnerability to climate change, it has been taking crucial steps towards climate adaptation and mitigation. The government has issued several adaptation policies, plans and agreements such as the 2012 National Disaster Risk Management Plan, the 2013 National

[7] World Bank Climate Change Knowledge Portal. Climatology | Climate Change Knowledge Portal. (n.d.).https://climateknowledgeportal.worldbank.org/country/kiribati/climate-data-historical

Communication under the United Nations Framework Convention on Climate Change, the Nationally Determined Contributions (NDCs) 2016, and the 2018 Kiribati Climate Change Policy. Five key adaptation focus areas have been highlighted including early warning systems, resilient infrastructure, protecting dryland agriculture crop production, mangrove planting, and making water resources management more resilient.

Maldives

Overview

Maldives is an archipelago of 26 low-lying coral atolls in the Indian Ocean located southwest of the Indian subcontinent. The country consists of just under 1,192 small tropical islands out of which about 358 are used for economic activities and human settlement. The driving force of Maldives' economy is tourism, which contributes about one-third of the GDP and is also the fastest-growing economic sector within the country. Though the contribution of fisheries and agriculture to GDP has been declining in current times, these sectors are still a major source of income and subsistence for many rural communities. Due to a combination of political, geographic, and social factors, Maldives is recognized as highly vulnerable to climate change impacts.

Current and Future Impacts

Maldives has quite low elevation and varied topography due to which any small changes in sea level could mean extensive land flooding or Even submersion. It poses a looming threat to homes and industries situated near the coastline. More than 90 of the inhabited Maldives islands already experience annual floods generated by storms far away. Corals are highly sensitive to changes in temperature and, as a result, the incidence of bleaching will increase in frequency and intensity with the projected rise in sea surface temperatures.[8] Another potential area of vulnerability is the recreational diving sector, which is threatened by environmental degradation, loss of reefs, and coastal erosion. Fish acts as a primary source of dietary protein among Maldivians, hence the effect of climate change on local and regional fishery systems will play a role in impacting their diets.

Mitigation

A few key national adaptation strategies and policies include the Disaster Management Act, Maldives Climate Change Policy Framework, and most recently the Strategic Action Plan 2019–2023 (SAP). SAP aims to strengthen and increase crop production to ensure the nutritional safety of communities, self-sufficiency and lessen imports in the country. It also aims to ensure the sustainable development of fisheries by incorporating modern fishery management principles. It believes in supporting local communities to increase their resilience. It also aims to integrate climate risk hazards and vulnerabilities to local development planning across various islands, to improve quality of life.

OTHER UN BODIES IN ACTION

United Nations Development Programme (UNDP)

UNDP has a strong hold and support over the SIDS regions which it uses efficiently to support these countries by enhancing climate action, propelling marine-based economies and catalysing a digital transformation. Through the Climate Promise initiative, UNDP is working with SIDS to prepare enhanced Nationally Determined Contributions (NDCs) that are goal-

[8] Maldives Climate Change Profile. (n.d.-f). https://climateknowledgeportal.worldbank.org/sites/default/files/2020-12/15649-WB_Maldives%20Country%20Profile-WEB.pdf

oriented and comprehensive. For example, the Cook Islands has achieved quite a lot of their emissions reduction targets and is now aiming for net zero emissions between 2030 and 2040.[9] Antigua and Barbuda is embarking on an equitable transition of its workforce with an energy target of 86 per cent renewable energy from local resources by 2030. In Comoros, it also created a data platform for SIDS, which is an open-sourced digital tool that provides revised, consistent, and comprehensive information on national and geospatial data, and other development metrics.

United Nations Children's Fund (UNICEF)

UNICEF has recognised that children in SIDS are particularly vulnerable, facing risks including poverty, health, food, and nutrition insecurity, child protection, educational disruptions, and displacement challenges. They support children and young people in SIDS with various mitigation strategies amidst climate disasters. They have fine-tuned their approaches towards a more youth-oriented system that is needed to address their vulnerabilities in SIDS. UNICEF has Responded to children's nutritional needs by promoting climate-resilient food systems, and healthy diets. They work in partnerships with governments to develop robust policies and advocate for children's rights in the global climate.

United Nations Environment Programme (UNEP)

UNEP has been a crucial contributing agency towards SIDS. UNEP along with UNFCCC and other UN organisations supported the 'SIDS Coalition for Nature' which is a platform created for SIDS members to advocate on agreed common SIDS priorities and needs. It is an important organising mechanism to unite SIDS members around the implementation of the Global Biodiversity Framework targets. UNEP has been supporting the development and implementation of legal and collaborative frameworks and strategic planning to address the rising concern of plastic pollution in several countries, including SIDS such as Mauritius, Seychelles, and Trinidad and Tobago. Additionally, a myriad of national and regional capacity development activities are designed, considering SIDS' unique circumstances and needs. It is also the lead agency of the Global Environment Facility (GEF) Islands Programme "Implementing Sustainable Low and Non-Chemical Development in Small Island Developing States" (ISLANDS) and is supporting more than 33 island nations including many SIDS in the Atlantic, Caribbean, Indian and Pacific regions to improve chemicals and waste management.[10]

World Bank

The World Bank Group has agreed to a Regional Partnership Framework (RPF) with Kiribati, Nauru, Marshall Islands, Federated States of Micronesia, Palau, Samoa, Tonga, Tuvalu, and Vanuatu where climate change is one of four key focus areas of the agreement. Under the heading of strengthening resilience to natural disasters and climate change, the RPF aims to continue to support regional and single-country activities that help them strengthen their resilience against natural disasters and climate change.

[9] Small island developing states are on the frontlines of climate change – here's why. UNDP Climate Promise. https://climatepromise.undp.org/news-and-stories/small-island-developing-states-are-frontlines-climate-change-heres-why

[10] Environment, U. (n.d.). UNEP & Small Island Developing States (SIDS). UNEP. https://www.unep.org/topics/ocean-seas-and-coasts/small-island-developing-states/unep-small-island-developing-states

They will ensure that at least 35 percent of the total portfolio will directly or indirectly support climate-related co-benefits. The RPF further identifies a range of regional and country-specific interventions including vulnerability assessment and disaster risk planning, financing and insurance initiatives for climate risks and natural hazards, as well as support to resilience-building interventions in areas such as transport, agriculture and water supply.

PAST RESOLUTIONS

Barbados Programme of Action- 1994

It was established by the UN Global Conference on the Sustainable Development of SIDS which was held in Barbados from 25 April to 6 May 1994. It created specific policies and actions for countries, regions, and international efforts. Their commitment to these goals is reflected in the Barbados Programme of Action (BPOA). The resolution focuses on areas such as climate change and sea-level rise, natural and environmental disasters, management of wastes, and coastal and marine, freshwater, and biodiversity resources. It highlights the special challenges and constraints that cause major setbacks to the socio-economic development of SIDS, including small size and geographic Isolation that prevent the scaling of their economies.[11]

SAMOA Pathway - 2014

The SAMOA Pathway expanded the mandate of the UN Office of the High Representative for the Least Developed Countries, Landlocked Developing Countries and Small Island Developing States (UN-OHRLLS) to include SIDS. It aims to address the unique challenges faced by SIDS and support the coordinated follow-up of the Programme of Action for the Sustainable Development of SIDS. It will undertake advocacy work in favour of the small island developing States in partnership with the relevant parts of the United Nations. It also aims to assist in mobilising international support and resources for the implementation of the Programme of Action. Earlier SIDS Member States were never able to adequately report their progress. So UN/OHRLLS undertook this exercise to develop a set of instruments intended to simplify the reporting process for SIDS. The approach has been to use the 'Framework of Monitoring Indicators for the SAMOA Pathway' as its foundational point to minimise overlapping efforts both by the UN and SIDS Member States.[12]

Paris Agreement - 2015

The Paris Agreement is a legally binding international treaty on climate change adopted By 196 Parties at the UN Climate Change Conference (COP21) in Paris, France, on 12 December 2015. Its primary goal is to hold "the increase in the global average temperature to well below 2°C above pre-industrial levels" and pursue efforts "to limit the temperature increase to 1.5°C above pre-industrial levels.".[13] It follows a five-year plan of ambitious climate actions which get more progressive with time. It includes Nationally Determined Contributions (NDCs) where countries communicate actions they will

[11] United Nations. (n.d.-e). BPOA - Barbados programme of action sustainable development knowledge platform. United Nations. https://sustainabledevelopment.un.org/conferences/bpoa1994

[12] United Nations. (n.d.-i). The Samoa pathway | office of the high representative for the least developed countries, landlocked developing countries and Small Island Developing States. United Nations. https://www.un.org/ohrlls/content/samoa-pathway

[13] *Paris Agreement*. Unfccc.int. (n.d.). https://unfccc.int/process-and-meetings/the-paris-agreement

take to reduce their greenhouse gas emissions to reach the goals of the Paris Agreement.

CONCLUSION

Small Island Developing States (SIDS) stand on the frontline of extinction due to the adverse effects of climate change, impacting both the biosphere and abiotic factors. These inhumane conditions pose significant threats to humanity as a whole. The destruction of income sources pushes these nations toward poverty, which must be avoided at all costs. It is equally important to address both terrestrial and marine ecosystems, as the uninhabitable conditions in water bodies further exacerbate the challenges faced by biotic creatures residing in them.

Understanding the severity of climate change's impacts on SIDS necessitates the implementation of feasible and viable solutions to mitigate these aftereffects. Each small, vigilant step taken toward overcoming these challenges can help ensure that the populations in these regions can thrive and pass on their traditions and cultures to future generations, securing a stable future.

Agenda 2: Advancing Climate Action in Global Food Security

AGENDA OVERVIEW

The agenda aims to bring to light the variety of climatic phenomena affecting food security at a global level. Food security is a basic right of every individual which many have been deprived of for centuries. In recent developments, through direct and indirect patterns, climate change has affected the availability and accessibility of safe and secure food commodities. Droughts, storms, rising sea levels, increasing global temperatures and greenhouse emissions are a few of the factors that have led to an increase in food insecurity, particularly in low-income countries that have a finite amount of resources to tackle this problem. It has increased malnutrition rates, particularly amongst children and contributed to the loss of the economy leading to economic and political instability across countries. By advancing climate action, it is essential to mitigate and contain the effects to a minimum extent. Only with the joint efforts of every country in the world, we can aim to accomplish our food target goals by this century.

Increasing Climate Resilience in Crops
Source: Foundation for Food And Agriculture Research

INTRODUCTION TO GLOBAL FOOD SECURITY

The World Food Summit defined food security as the state in which people at all times have physical, social and economic access to sufficient and nutritious food that meets their dietary needs for a healthy and active life. While there are a myriad of factors affecting food security such as economic and political instability, access to resources, social inequality and others, climatic phenomena play their due part in the process. Global warming is influencing weather patterns, causing heat waves, heavy rainfall, and droughts. This extreme weather causes disruptions in crop growing and over time changes the regional growing conditions.

Low-income and developing countries in Sub-Saharan Africa, and South and Southeast Asia face challenges related to food security due to climate change. Most recent ongoing incidents have been observed in African countries including Ethiopia, Niger, and Mali where the extreme weather patterns ranging from droughts to floods across the country have driven the countries' vulnerable populations into severe starvation. As the frequency of these events has been rising, aid from other nations has struggled to keep pace with the growing challenges. These countries are economically dependent on agriculture and farming hence they are not only affected by food insecurity but also by an immense financial crisis.[14] These issues can be resolved by implementing various technological and policy solutions by the governments. With the right global efforts, we can develop resilient adaptation and recovery.

CAUSES

Topographical Factors

Loss of fertile lands

The area of cultivable farmland has experienced a stark decline, driven by climatic phenomena that are accelerating desertification. An increasing expanse of land has been lost to diminished fertility. It is estimated that today about 500 million people live in areas that are experiencing desertification.[15] Diverse climatic factors critically undermine soil quality, leading to the depletion of essential nutrients. Hence the soil's capacity to support the cultivation of high-quality crops at necessary rates is severely compromised, resulting in a significant decrease in food production.

Soil Erosion

The rising frequency and intensity of climatic disasters—such as prolonged droughts and extreme precipitation—increases the degradation of the upper soil layers. The nutrient-rich topsoil gets eroded, significantly impeding the land's ability to sustain agricultural productivity. This soil also experiences a decline in its water retention capabilities leading to increased surface water runoff which in itself is a cause of further soil erosion. As a result, crops do not receive the necessary quantity of water for optimal growth, leading to a decline in their overall quality. This situation heightens the demand for fertilisers, creating a vicious cycle of land degradation.

Agricultural Impacts of Climate Crisis
Source: IOFS

[14] World Bank Group. (2022, October 19). Climate explainer: Food Security and climate change. World Bank. https://www.worldbank.org/en/news/feature/2022/10/17/what-you-need-to-know-about-food-security-and-climate-change

[15] United Nations. (n.d.-j). World food security increasingly at risk due to "unprecedented" climate change impact, New Un report warns | UN News. United Nations. https://news.un.org/en/story/2019/08/1043921

Marine-Based Influences

Rising Sea Levels

Global mean sea level has been reaching record high levels after increasing at an average of 4.5 mm per year since 2013. This is more than double the rate between 1993 and 2002 and is mainly due to the accelerated loss of ice sheets in the Arctic and other poleward regions. Rising sea levels lead to increased salinity of our oceans and seas. This heightened salinity poses a significant threat to agricultural lands, which are often strategically located near these water bodies for easy access to irrigation resources. As flooding events become more frequent, farmlands are flooded and harvests are devastated.

Scarcity of Freshwater Resources

The saline water seeps into the surrounding groundwater, causing saltwater intrusion and making freshwater sources contaminated and unsuitable for agricultural use. The deterioration of these freshwater resources undermines crop quality and yields, affecting food security. Access to freshwater resources has diminished by 20 percent over the past two decades, significantly limiting individuals' ability to secure safe water essential for survival. In the absence of clean water, the agricultural crops cultivated may also be contaminated, making them unsuitable for consumption and harming public health. This decline not only threatens food security but also poses serious risks to the well-being of communities reliant on this vital resource.

Natural Calamities

Impact of Disasters on Food Security
Source: UN Water

The increasing frequency and intensity of extreme weather disasters such as floods, droughts and forest fires due to climate change are having a devastating effect on food security and livelihoods. According to a report released by the Food and Agriculture Organisation (FAO), the annual occurrence of disasters is now more than three times that of the 1970s and 1980s as a result of our warming climate. More than 63 percent of the repercussions are borne by the agricultural sector with the least developed and low- and middle-income countries bearing a significant toll of these issues.[16] The report has stated drought to be the single most adverse effect that causes agricultural production loss, followed by floods, storms, pests and diseases, and wildfires. It has been estimated that the annual loss of crop production due to natural disasters could fulfil the necessary food intake of more than seven million adults annually. Hence addressing these challenges is crucial for safeguarding against loss of food supplies for the people.

Atmospheric Factors

Greenhouse emissions are one of the biggest contributors to climate change accounting for almost one-third of the world’s climate change effects. Hence it also tends to be a major contributing factor towards food insecurity. In

[16] Climate change Related Disasters. Unfccc.int. (n.d.-a). https://unfccc.int/news/climate-change-related-disasters-a-major-threat-to-food-security-fao

recent times as close as 2023, carbon dioxide emissions were recorded as over 420 parts per million which is way beyond the 'safe' limit of 350 parts per million of carbon dioxide in the atmosphere. For example, global yield trends of both maize and wheat are expected to show a significant decline in future projections. These declines can be attributed to the negative impacts of climate change arising from increasing greenhouse gas emissions. Maize and wheat are a major staple food ingredient in the diets of communities in many less developed countries of Africa and Central America. The consumption of these grains crosses more than 700 million metric tonnes annually which provides for more than 20 percent of the world's calorie and protein intake. Hence the only viable method to sustain food security is via increasing food production while making an effort to reduce our carbon emissions at the earliest.

IMPACTS

Human Health

Major influences undermining nutrition include reduced crop yields and subpar quality of crops lacking nutrients. Contaminated water and improper sanitation may also be a contributing factor. By 2050, the risk of hunger and malnutrition could rise by 20 percent.[17] Malnutrition will claim the lives of countless young children if global actions fail to reach their goals. Climate shocks are projected to increase everyday burdens on women along with negative impacts on the care of children. Due to drought and desertification, women have to redirect their attention from childcare to search for water and firewood for their livelihoods and sustenance. This proves to be harmful to the women's physical as well as mental well-being and the child might not receive adequate care for healthy growth. Temperature and precipitation changes enhance the spread of vector-borne diseases. Without preventive actions, deaths from such diseases may rise well over 7,00,000 annually. Human health remains consistently vulnerable due to the lack of such basic food amenities and the constant threat of disease.

Social Discrimination

The populations at greatest risk are those that are dependent on agriculture and natural resources, with livelihoods that are highly exposed to climate change impacts. In regions with high levels of food insecurity and inequality, increased frequency of droughts will affect poorer households. Even amongst these households, as women have higher vulnerability and restricted access to resources, they will be predominantly affected. Indigenous peoples, who depend on the environment and its biodiversity for their food security and nutrition, are at high risk, especially those living in areas where significant impacts are expected such as The Arctic, mountain areas, the Pacific islands, coastal and other low-lying areas. This leaves them with the sole option to migrate to other regions which will bring about another wide range of implications.

Industrial Loss

Crop Production

Climate change has both direct and indirect impacts on agricultural production systems. Apart from the obvious direct effects such as an increase in temperature levels, or rainfall patterns, indirect impacts also play a major role. These include changes in other species such as pollinators, pests, disease vectors and invasive species. Pests and diseases are likely to move in

[17] Climate crisis and malnutrition - a case for acting now. UN World Food Programme. (n.d.). https://www.wfp.org/publications/climate-crisis-and-malnutrition-case-acting-now

the direction of climate change. Consequently, previously immune areas are exposed to diseases they are unprepared for and hence do not have measures to manage and control them with potentially enhancing negative impacts. Multiple reports believe that the direction of yield will change in many major agricultural regions at both low and high latitudes, with strong negative impacts especially at higher levels of warming and at low latitudes. Four major crop groups of grains, oil seeds, wheat and rice will have a reduction in yield by at least 17 percent globally. This is particularly threatening considering these four crop crops account for more than 70 percent of global crop harvest.

Disruption in the Crop Production
Source: World Economic Forum

Livestock and Forest Resources

The most grave impacts are experienced in animal productivity and health as well as yields of forages and feed crops. In various countries in sub-Saharan Africa, 20 to 60 per cent losses in animal numbers were recorded during serious drought events in the past decades. In South Africa, dairy yields may decrease by 10 to 25 per cent because of climate change. Increased temperatures and reduced precipitation can cause important drops in fodder production.

In various regions, climate change is contributing to decreased productivity and waning of trees from drought and temperature stress, increased wind and water erosion, increased storm damage, increased frequency of forest fires, saltwater intrusion and sea level rise, and damage from coastal storms. This hampers forests from contributing to the resilience of agricultural systems, such as regulating the water and temperature at the landscape level and the provision of habitats for a variety of species.

Fisheries

Climate change affects commercial fisheries and the development of aquaculture in marine and freshwater environments. This is a result of an increase in temperature, storm systems, acidification, or an increase in salinity content. Incidences of coral reef bleaching have been on the rise, threatening the extinction of habitats of about 25 percent of marine species. Reorganisation of global marine fish is forecast on a huge scale, with a decrease of up to 40 per cent in the tropics, and an increase of 30 to 70 per cent in high-latitude regions.[18] In the Mediterranean, many invasive species have arrived from lower-latitude regions at an alarmingly fast rate. Riverine fish and their abundance and diversity are particularly sensitive to disturbances in the quantity and timing of water flows. Hence this will have a profound impact on the coastal communities who depend on the vital water bodies as a source of income and livelihood.

Economic Loss

At the farm or household level, climate change can lower income and stability by affecting productivity rates and prices. Farmers are often forced to sell off their productive assets, like

[18] Climate change and food security: Risks and responses. (n.d.-b). https://openknowledge.fao.org/server/api/core/bitstreams/a4fd8ac5-4582-4a66-91b0-55abf642a400/content

cattle, making them lose any long-term investments that might help them to survive. Poorest farmers will be affected most by weather-related income changes. At the national level, climate risks can lead to disruptions in agricultural production and food availability, potentially curtailing market stability and affecting supply and storage systems. This can drive up agricultural commodity prices, making food less accessible and stable for the population, especially in countries where many people spend a large portion of their income on food. Globally, climate change impacts supply chains and can cause food price spikes, leading to market volatility, changes in import/export behaviour, and disruptions in trade patterns.

BARRIERS

Geopolitical Tensions

Political instability

Climate change can increase pressure on governmental organizations and institutions, hindering their ability to provide essential services to the public. This often fuels mistrust and grievances among the populace, leading individuals to seek support from illegal organizations for survival. The loss of livelihoods leaves populations in desperation, making the promises of protection, income, and justice offered by such groups more appealing, while the true agendas of these groups often go unnoticed.

In the Lake Chad Basin region, Boko Haram has recruited members from local communities who are apprehensive towards the government due to a lack of economic opportunities and access to essential resources. Environmental degradation enables terror groups to extend their influence and manipulate resources to their advantage. In Iraq and Syria, water shortages have been exploited, with terror groups illegally controlling water infrastructure to impose their will on communities. Climate change has a multiplier effect, potentially instigating further instability, conflict, and terrorism.[19]

The compounded effect of these factors undermines a nation's food security. The inability to implement effective measures to counteract the adverse effects of climate change can be used to convince the public of the government's incompetence.

International conflicts

Climate change is an indirect factor influencing conflict, potentially worsening it under certain conditions. With the intensification of the climate crisis in the future, migration of people away from their homes towards search of secure circumstances will develop at alarming rates. This will be a result of everything ranging from desertification to rising sea levels. The groups most vulnerable under these cases include those who had to flee their country due to conflict and are now feeling the effects of a hostile climate. These people have little to no chance of ever returning to their homelands.

One of the most gut-wrenching images of the environmental cost of conflict was during the first Gulf War. This was when 700 of Kuwait's oil fields were set ablaze and the smoke plume above them initially stretched for 800 miles. An international coalition of firefighters bravely tackled the fires for months, losing many valuable lives until the last well was finally capped in November 1991. The horrifying effects of this disaster are still felt today with more than 90 per cent of the contaminated soil still

[19] United Nations. (n.d.-f). Climate change "a multiplier effect", aggravating instability, conflict, terrorism, secretary-general warns Security Council | Meetings Coverage and press releases. United Nations. https://press.un.org/en/2021/sgsm21074.doc.htm

exposed.[20] Some more such instances include the negative effects of the mass migration of refugees on the environment and the desertification of lands rendered uninhabitable by the bombings and landmines.

A common observation is that during political crises, the scarcity of resources will intensify and cause large-scale food shortages. Climate change adds to this issue, with its long-lasting and far-reaching impacts further straining agricultural systems and resource availability. The interconnection of conflict and climate disruption creates an unstable situation, where the challenges of securing sufficient food become increasingly intimidating and challenging.

Humanitarian Aid and Access

Funding in climate-vulnerable countries remains one of the most significant challenges faced globally. Over a decade ago, high-income nations pledged US$100 billion annually in climate finance to low-income countries. However, this long-standing commitment has yet to be fulfilled, with little indication of imminent progress. Several issues have been identified, including shortfalls in delivery, opacity in donor accounting, access problems, and the increasing prevalence of loans in the funding portfolio.[21] Given that developed countries contribute the most to the climate crisis while bearing the least of its impacts, it is incumbent upon them to lead the mobilisation of funds to support poorer nations in addressing this crisis. Beyond the absence of adequate financial aid, the lengthy and rigid process of securing funding presents a significant hurdle. The timeline from project formation to submission and approval can extend from one to two years, delaying critical assistance to affected areas. Additionally, the lack of flexibility in financing grants hampers the ability to address the urgent needs of developing countries effectively.

Ecological Imbalance

Climate change plays a role in the decline of biodiversity in ecosystems. Climate change has altered marine, terrestrial, and freshwater ecosystems across the globe. It has caused the loss of local species, increased diseases, and driven mass mortality of plants and animals, resulting in the first climate-driven extinctions. This leads to a stark decline in biodiversity. This gives rise to unsuitable conditions for crop cultivation in proximity to water bodies. It often results in diminished food production, compelling farmers to relinquish their lands. The health of these ecosystems is also affected due to influential shifts in the distribution of plants, animals, and human settlements. This increases the unwanted chance of spreading diseases via animals which can spill over to humans too. The rise in diseases adversely impacts the quality of crops cultivated. The contaminants such as excessive chemical use, and polluted water resources may cause the crop to pose health risks to humans.

Trade Disruption

Climate change effects such as extreme weather conditions raise the cost of trade by damaging the transport infrastructure and posing a challenge in the exchange of agricultural produce between nations. It can also cause productivity losses, supply shortages and transport disruptions, severely impacting trade. Export growth of agricultural products from least developed countries has been found to decrease

[20] Climate change. Unfccc.int. (n.d.-b). https://unfccc.int/news/conflict-and-climate

[21] Climate Finance and the USD 100 billion goal. OECD. (n.d.). https://www.oecd.org/en/topics/sub-issues/climate-finance-and-the-usd-100-billion-goal.html#:~:text=Share-,About,actions%20and%20transparency%20on%20implementation

by 2 to 5.7 per cent with a rise in the country's temperature of every 1°C. Maritime transport is under particular threat by climate change as it accounts for more than 80 per cent of world trade. Small economies and landlocked countries, which trade through a limited number of ports and routes, can suffer major trade challenges from climate-related disasters. As Agriculture is the most vulnerable sector with minute changes in temperature or precipitation it is raising serious concerns about future food security.

For instance, the Paraná River transports 90 per cent of Paraguay's international trade of agricultural goods, but recurrent droughts have in recent years frequently lowered water levels, causing congestion and delays as it decreases the volumes of goods that can be transported through. Sub-Saharan Africa and South Asia having a high share of agricultural employment may face more severe labour market disruptions and agricultural yield shocks over other regions. Another example is the Panama Canal which transports over 40 million tons of goods—constituting about 5 percent of global maritime trade. However, the worst drought in the canal's 143-year history has caused water levels to recede, significantly limiting ship passage and disrupting global trade markets.[22]

Status Discrimination

In recent decades, global economic growth has been on a steady rise, lifting millions out of extreme poverty and reducing inequalities between countries. However, this hard-earned progress is under threat of reversal due to the unmanaged climate crisis, which disproportionately affects the poorest regions and populations. This situation risks increasing inequality between countries once again. The impacts of climate change also threaten to exacerbate inequality within countries. In the poorest economies, a large portion of the population relies heavily on the agricultural, forestry, and fisheries sectors, which are among the first to be affected by climate change.

Since these individuals depend on nature for their livelihoods, they are consistently the most impacted. Rising sea levels and increasing groundwater salinity worsen pre-existing disparities in access to clean water and affordable food. The most marginalised populations often lack access to essential mitigation resources, financial reserves, and technological innovations that could help rebuild their livelihoods. As a result, they are particularly vulnerable to the devastating impacts of climate disasters, which can completely uproot their lives with little hope of recovery.

Rapid Urbanisation

Rapid urbanisation is making people more vulnerable to the impacts of climate change. More than half the world's population today live in cities, and another 2.5 billion people are expected to join them by 2050.[23] Densely populated cities like New York, Mumbai and Jakarta have seen a massive rise in the intensity and frequency of torrential rain, where the worst is borne by the people living in isolated and marginalised communities such as the slums. This causes a rise in socioeconomic disparities, leaving communities in slums with insufficient access to vital food resources. The ever-increasing population also elevates the demand for food supplies; however, the conversion of arable land

[22] Arslanalp, S., Koepke, R., Sozzi, A., & Verschuur, J. Climate change is disrupting global trade. IMF. https://www.imf.org/en/Blogs/Articles/2023/11/15/climate-change-is-disrupting-global-trade

[23] UNFCCC. int. (n.d.-d). https://unfccc.int/news/rapid-urbanization-increases-climate-risk-for-billions-of-people

into housing for urban residents poses a significant challenge in meeting these needs.

CASE STUDIES

Columbia

Ranking 64th on the Global Food Security Index 2022, Colombia shows a slight improvement by having a decrease of 5 percent in food insecurity, with 13 million people facing moderate or severe food insecurity - down from 15 million the previous year.[24] Of the 15.5 million Colombians who are food insecure, 2.1 million are severely food insecure and 13.4 million are moderately food insecure.

Although there is an improvement, half of the Colombian households are in a situation of marginal food security and are continuing to be at risk due to external shocks. The major impacts faced are because of El Nino and La Nina which gives rise to variations in water flows and the risk of forest fire. Additionally, this causes food prices to show an unstable trend which can deeply affect the citizens, more specifically the poorer section. Even if there has been an increment in the household capacity to purchase food since the year 2022, according to a report in February 2024, 43 percent of households are still facing severe difficulties in accessing food since last year.

WFP has taken the responsibility of working hand in hand with several institutions and organisations to help mitigate the solution and address hunger and poverty through development programs. Some of the initiatives taken by the WFP are training, seed funding, and technical assistance.[25]

Finland

Ranking 1st on the Global Food Security Index 2022, Finland has almost the perfect desired results in all the categories that label a nation as food secure. The prevalence of undernourishment is 2.5 percent which is very impressive considering the statistics of other nations.

Finland has the goal to achieve a net-zero climate target by 2035 and soon after achieve the lowest levels of carbon emissions.[26] Additionally, there has been a target to cease biodiversity loss by 2030. Even though there are targets set in different directions, the main priority is given to food security. Continuous efforts are being put in to ensure that the relatively high level of food security and self-sufficiency in food production is maintained. They also have aimed to develop the region and local-based food systems to distribute the production and to mainly affirm that every citizen gets ample amounts of food resources for survival.

The nation is very supportive of developing countries' agricultural growth to improve worldwide food security. The Finnish also support and promote agricultural research and wholeheartedly support farmers' organisations.

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[24] Colombia sees a modest improvement in food security but half of the population remains exposed to climatic and economic risks: World Food Programme. UN World Food Programme. (n.d.-b). https://www.wfp.org/news/colombia-sees-modest-improvement-food-security-half-population-remains-exposed-climatic-and

[25] Colombia: Productive and food secure territories for a peaceful and resilient Cauca. Sustainable Development Goals Fund. https://www.sdgfund.org/case-study/colombia-productive-and-food-secure-territories-peaceful-and-resilient-cauca

[26] *Finland*. Home - Food, Agriculture, Biodiversity, Land-Use, and Energy (FABLE) Consortium. (n.d.). https://fableconsortium.org/finland/

Furthermore, Finland helps in mitigation of and adaptation to climate at all levels– international to grassroots levels. [27]

PAST INTERNATIONAL ACTIONS

Food and Agricultural Organisation (FAO)

An organisation specifically focused on food security was established at the United Nations in 1945. With the goal of achieving a world free from hunger, the FAO has developed a wide range of resolutions. One of its recent strategies involves looking beyond food production to encompass domains such as livestock, ecosystems, and biodiversity. This strategy aims to build resilience against climate change in the context of food security and aligns with the principles of the Paris Agreement, with a tentative deadline of 2031. The optimal solution derived from various discussions to address climate change is the transformation of agrifood systems

United Nations Environment Programme (UNEP)

The United Nations Environment Programme worked towards understanding the linkages between climate change and security. The term security included food and water shortage and competition over natural resources. The international report "A New Climate for Peace" published in 2015 aimed to discuss seven key compound climate and fragility risks that needed immediate action. The volatile food prices and provisions lead to numerous cases of starvation and malnutrition throughout the world.[28]

World Food Programme (WFP)

The vast majority of people in the world face hunger as a consequence of climate change. WFP focuses on combating it by ensuring that food-insecure communities can prepare for, face and recover from the climatic shocks and operations. WFP also implements climate risk management solutions in 42 countries, benefiting more than 15 million people. Food security begins from the roots which is ensuring that disaster-stricken communities can bounce back with proper restoration of the agricultural systems. Up to 2021, WFP has rehabilitated 1.6 million hectares of degraded land, built 111,000 water ponds and planted 60,000 hectares of forests. Simultaneously, WFP helps in developing early warning systems to trigger humanitarian action before the climate change impacts are even felt as this could save them from major damage and could help in preparing to combat the same. [29]

PAST RESOLUTIONS

Plan of Action on World Food Security - 1996

The World Food Summit under the guidance of the FAO and the UN initiated the idea of taking immediate action against the imbalance in the rights of humans to have access to proper hygienic and healthy food. It is the basic right of every human to be free from hunger yet in reality, the circumstances are much different and infeasible as people around the world are denied

[27] Food security and natural resources. (n.d.-e). https://um.fi/documents/35732/48132/food_security_and_natural_resources__including_access_to_water_and/dd7430a1-a30e-1f8c-129f-cd9c8e5e9c91?t=1525690524377

[28] United Nations. (n.d.-f). Climate action can help fight hunger, avoid conflicts, official tells Security Council, urging greater investment in adaptation, resilience, clean energy | meetings coverage and press releases. United Nations. https://press.un.org/en/2024/sc15589.doc.htm

[29] Climate action: World Food Programme. UN World Food Programme. (n.d.-a). https://www.wfp.org/climate-action

their right due to their economic shortage. The man-made and natural disasters affect the growth of this produce and lead to a shortage in food supply. The plan ensures protection against these situations by being best prepared and able to meet transitory and emergency requirements with proper recovery, rehabilitation, development and stock enough to fulfil further needs. [30]

United Nations Millennium Declaration - 2000

The effective 3-day summit held from 6th to 8th September 2000 in the United Nations filled with global leaders gave rise to the Millenium Declaration which was the main documentation inclusive of the goals, values and principles to be achieved and followed on an international scale for the twenty-first century. It had shown emphasis on all the basic human rights and privileges that had to be imparted to every citizen globally. In addition to a key focus on globalisation, it also highlighted the need for the eradication of hunger. The domain of undernutrition covered foetal growth restriction, stunting, wasting and deficiencies of vitamin A and zinc, along with suboptimal breastfeeding. Ultimately, this is one of the general causes of death in an estimated 45 per cent of all deaths among children of age under 5 years. The targets were set to be achieved by 2015. A major achievement accomplished included the proportion of underweight children in developing countries decreased from 28 per cent to 17 per cent between the years 1990 and 2013. Although this enhancement seems promising, the only flaw is that the distribution of the improvement is imbalanced between and within the regions.[31]

CONCLUSION

Global food security has been one of the more pressing concerns of this century. For decades it has faced constant threats from various political, and economic factors but in recent years climate change has emerged as a major driver of food insecurity. Climate impacts agricultural productivity, food availability and livelihoods, particularly in low-income countries. In these regions immediate action is crucial. Policymakers across the globe must unite to address this critical issue and mitigate the disruptions caused by climate change. This will require significant investments towards innovative technologies and infrastructure to support those most affected. Fostering equitable distribution of resources and inculcating more sustainable practices must become standard practice. Not just for the current generations but for the future ones as well, we must build more climate-resilient food systems. Our singular goal must be to ensure a future where access to food is a fundamental right, not a privilege, even in the face of the climate crisis.

[30] Rome declaration on world food security. Rome Declaration and Plan of Action. (n.d.-b). https://www.fao.org/4/w3613e/w3613e00.htm#:~:text=1.,national%2C%20regional%20and%20global%20levels

[31] United Nations. (n.d.-m). United Nations Millennium Declaration. United Nations. https://www.un.org/en/development/devagenda/millennium.shtml